

Confidence Signals at the MiniMax Frontier

A living research programme for difficulty-response curves

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Status: ongoing, preliminary, and not peer reviewed. This page is a research programme, not a finished paper. It records what is exploratory, what has been specified prospectively, and what evidence would change the current view.

The question

Difficulty-response curves estimate the probability that a model succeeds as a task gets harder. The next question is whether information produced during a single call can improve that probability:

Can reasoning traces, billing metadata, latency, or throughput improve a calibrated probability of correctness beyond the difficulty curve itself?

MiniMax-M2.5 is the first deep dive. Its existing visible-trace frontier lane contains correct answers, silently wrong answers, and conspicuous truncations. That makes it possible to distinguish two operational problems that are often blurred together: detecting an answer that is wrong despite completing normally, and detecting a call that failed loudly.

Current evidence

Status	Claim	Current reading
Supported	Difficulty curves provide a useful baseline probability.	The existing instrument estimates a frontier and sharpness from fresh, mechanically graded tasks.
Exploratory	MiniMax trace texture ranks silently wrong calls.	A previously held-out trace judge reached AUC 0.93, but its false-alarm rate was 31%. This is a lead, not a calibrated-confidence result.
Exploratory	Billing and serving metadata add information at the frontier.	Existing calls record billed tokens, latency, attempts, route, finish reason, and correctness. Their incremental value over the curve has not yet been measured prospectively.
Prospective	A frozen combined score improves probability calibration.	The MiniMax v1 protocol specifies the comparison, endpoints, stopping rule, and spend gate before any new calls.
Censored	End-to-end billed tokens per second measures generation speed.	It does not: the denominator includes queuing, retries, transport, and provider overhead. Streaming telemetry is required for an observed output rate.

Study sequence

1. **Retrospective calibration.** Treat all existing MiniMax calls as exploratory for this new question. Compare a curve-only prior with metadata-only, trace-only, and combined confidence models using instance-grouped out-of-sample predictions.
2. **Spend gate.** Spend nothing unless the retrospective cohort contains at least 20 silently wrong calls, the best frozen model reaches AUROC 0.75, and its Brier skill over the curve prior is at least 10%.
3. **Prospective validation.** If the gate passes, evaluate the frozen score once on fresh instances. Stop after at least 60 correct and 60 silently wrong completed calls, or at 600 calls / USD 40.
4. **Streaming feasibility.** Reserve at most USD 10. Proceed only if the prospective confidence result passes. Measure time to first token, reasoning and answer channel timing, inter-chunk gaps, and observed output rate.

The complete frozen protocol is MiniMax confidence v1.

Why this differs from a confidence heuristic

The primary endpoint is the Brier score: the mean squared error of the stated probability. A useful score must improve probability calibration, not merely rank failures. AUROC, log loss, calibration plots, and risk-coverage curves are secondary views. All folds are grouped by generated instance so repeated samples from the same puzzle cannot leak across training and evaluation.

The model, route, provider, token cap, prompt version, and frontier window are fixed. Correct completed answers and silently wrong completed answers form the confidence task. Truncations, parse failures, refusals, timeouts, and API errors are a separate loud-failure outcome.

Related work

This programme sits between dynamic model evaluation, psychometric model measurement, and inference-time routing. Generated evaluations such as GSM-Symbolic and Dynamic Evaluation by Meta Probing Agents reduce dependence on static test banks. IRT-Router and RouteLLM connect estimated difficulty or preference to model selection. Self-Consistency uses agreement across sampled reasoning paths. The present study asks a narrower deployment question: after controlling for a model's measured difficulty frontier, does its own trace or serving telemetry provide calibrated evidence about this particular answer?

Open record

The dated field journal is preserved as process evidence. The public protocol is committed before the prospective batch. Results are reported whether the gate passes or fails, and a failed gate stops spend rather than becoming a new round of feature search.

Code is licensed under MIT. Research text and released study data are licensed under CC BY 4.0, subject to the terms of the providers that produced model responses.